

Arizona State University
Department of Economics
Ph.D. Qualifying Exam

Macroeconomic Theory

June 1, 2021

Instructions: Read the questions carefully, show all your work, and explain all your answers. Indicate carefully if you introduce notation, or make assumptions when answering a question. Please make sure to allocate time to ALL three problems.

Good luck!

1. This question is about participation and employment in two-person households. Consider the following version of the overlapping generations model with capital. Time is discrete ($t = 0, 1, 2, \dots$). Individuals are born into households with *two* individuals. They live for two periods, and household members can choose to work in only the first period of their lives. For simplicity, there is no population growth.

Households value joint consumption in both periods, and dislike *joint* work (i.e. a case in which both individuals work). They are heterogenous in their distaste for joint work according to the realization of the variable q at the start of life. q is distributed across all newborns with cdf $G(q)$, and density $g(q)$. Preferences are represented by

$$U(c_{1,t}, c_{2,t+1}, q) = 2[\ln(c_{1,t}) + \beta \ln(c_{2,t+1})] - \iota q,$$

where ι is an indicator function that equals 1 if both individuals work and 0 otherwise.

In each household, one individual is endowed with 1 unit of efficiency units of labor while the other is endowed with $h \in (0, 1]$. All efficiency units are perfect substitutes in production. Competitive firms have access to a constant-return-to-scale technology that uses capital (K) and labor services L via the production function $F(K, L)$. The function F is strictly increasing in both arguments, strictly concave, differentiable and satisfies the standard Inada conditions. For simplicity, assume that there is no growth in labor efficiency.

- (a) Formulate the problem of a young household at date t . Show that there exists a unique threshold value q^* , such that for households with $q \leq q^*$ both members work, and for all households with $q > q^*$ only one household member works. Is q^* time invariant? Explain.
- (b) Show that q^* is strictly increasing in h . Explain.
- (c) Assume from now onwards that the production technology is Cobb-Douglas. Derive the law of motion for the capital to labor ratio. Argue that there is a unique, steady-state capital to labor ratio.
- (d) Consider two economies, A and B . These economies differ in the efficiency units of secondary earners, h , with $h^A > h^B$. Given this observation, economist X states that in steady state, output and the participation rate of secondary earners will be higher in A . Economist Y argues further that rental rate of capital in steady state in A will be lower. Are X and/or Y correct? Explain carefully.

2. Consider the following version of the Lucas tree model. There is a single divisible asset (“tree”) in fixed supply. Its dividends follow a Markov process with the transition function Q . There is a continuum of measure one of identical households with preferences over consumption given by

$$E_0 \sum_{t=0}^{\infty} \beta^t U(c_t),$$

where c_t is consumption at time t , function U is strictly increasing, strictly concave, C^2 , and $\beta \in (0, 1)$. Households are initially endowed with one unit of the asset.

The government taxes households’ income in each period at rate $\tau \in [0, 1)$, where income for tax purposes is defined as dividends plus the value of assets owned at the beginning of the period. Tax collections are distributed among households through lump-sum rebates. You are asked to analyze a recursive competitive equilibrium in which the equilibrium price function is stationary.

- (a) Write down the problem of the representative household in a recursive form. Derive the first-order and Envelope conditions. Derive the stochastic Euler equation evaluated at the equilibrium allocations and prices.
- (b) Using recursive methods, prove that the equilibrium asset prices in all states are strictly decreasing in the tax rate τ . Explain this result intuitively.
- (c) Suppose that the utility function is logarithmic, $U(c) = \ln(c)$. Find the closed-form expression for the equilibrium asset prices.
- (d) Economist A argues that due to distortionary taxes, the competitive equilibrium is inefficient. Is he right? Explain your answer.

3. Consider a discrete-time infinite horizon decision problem of an individual who decides how much to invest in skill accumulation in every period. If in period t the individual has skill s_t , she supplies s_t units of effective labor to the labor market, for which she is compensated at rate w (constant across periods). The labor income can be spent to finance consumption c_t in period t or to invest i_t in skill accumulation. The law of motion for the individual's skill is given by

$$s_{t+1} = (1 - \delta)s_t + As_t^{1-\gamma}i_t^\gamma, \quad t \geq 0,$$

where $\delta, \gamma \in (0, 1)$ and $A > 0$. The individual's life-time utility is given by

$$\sum_{t=0}^{\infty} \beta^t u(c_t),$$

where $\beta \in (0, 1)$ and $u : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ is a strictly increasing, strictly concave, and continuously-differentiable function, such that $\lim_{c \rightarrow 0} u'(c) = +\infty$.

No storage technology is available, and the individual has no access to asset markets.

- (a) Set up the individual's decision problem recursively. Derive the Euler equation and interpret it intuitively.
- (b) From now onwards, assume that $u(c) = \ln(c)$. Using a “guess and verify” approach, show that the individual's optimal investment choice is linear in her skill.
- (c) How does the individual's optimal investment in skill vary with δ ? Provide a formal argument and an intuitive interpretation.